

Do LLM Forecasts Change Their Minds Coherently?

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Abstract

Forecasting benchmarks usually score where a model finishes, not how it gets there. We introduce a martingale audit for the full probability path. For a coherent forecast, expected squared movement matches expected uncertainty reduction; their difference gives an intermediate-label-free residual $C = M - R$. Across 73 resolved markets, the eight-model primary panel has a model–crowd coherence gap of +0.061. The result survives alternative prompts, observation grids, missing-path treatments, and a deliberately noisier repeated-elicitation stress test. It reappears on 41 prospectively collected resolved markets and is concentrated in efficient systems, while the strongest systems remain close to the crowd. The model and crowd have different information sets, so the result is comparative rather than causal. Even under that constraint, endpoint scoring alone misses a systematic difference in how LLM forecasts evolve as evidence arrives.

Introduction

Forecasts are usually judged by their final answer. That misses how they behave while events unfold. A model may finish near the right probability after large, unstable revisions, or look stable because it barely reacts to new information. Recent benchmarks increasingly evaluate models on live questions, prediction markets, and replayed evidence [1–5].

Martingale and coherence tests have also been used to study probabilistic behavior in LLMs. Falck et al. [9] test in-context learning on exchangeable synthetic data; He et al. [10] measure whether expressed belief updates are predictable from prior beliefs during reasoning; and Kim et al. [11] study drift under repeated answer resampling. Paleka et al. [12] test logical and arbitrage consistency across related forecasting questions, while Andrews and Sarkar [13] construct outcome-free Dutch-book audits. Our target is different: temporal coherence within one forecast as dated evidence arrives, scored by squared movement versus uncertainty reduction and compared with the crowd on the same market.

For a binary event, the expected sum of squared probability changes equals the expected reduction in uncertainty [14]. We use this identity to ask whether model paths carry more movement than their own uncertainty reduction explains. The paired gap is positive, survives prospective replication, and falls sharply across the fixed capability tiers. The contribution is not another endpoint benchmark, but a compact diagnostic for how forecasts change.

How the audit works

Score and estimand. For probabilities $p_0, \dots, p_K$ observed on a fixed grid, define

$$\begin{aligned} M &= \sum_{k=1}^K (p_k - p_{k-1})^2, \\ R &= p_0(1 - p_0) - p_K(1 - p_K), \\ C &= M - R. \end{aligned} \tag{1}$$

For a square-integrable martingale, $\mathbb{E}[C] = 0$. Positive C means the realized path moved more than its uncertainty reduction; it does not by itself prove incoherence. The primary market-level comparison is

$$\Delta C_i = \frac{1}{J_i} \sum_{j=1}^{J_i} C_{ij}^{\text{model}} - C_i^{\text{crowd}}. \tag{2}$$

The residual C requires no gold intermediate probabilities. The paired ΔC uses the crowd as a benchmark, scored by the same identity rather than treated as ground truth.

Scope of the identity. Let $\mathcal{F}_0 \subset \dots \subset \mathcal{F}_K$ denote the information available at successive cut-points. The model receives nested snapshots: dated headlines are appended but never removed, and it sees neither prices nor prior model outputs. The crowd has a broader, different filtration that includes the market’s information flow. Our inferential target is the mean residual over the market and topic population represented by the study. Markets are the paired units and topics are the clustering units. We make no pathwise or universal model-population claim.

Empirical null. For market i , the null interprets each elicited value as $p_{ik} \approx \Pr(Y_i = 1 \mid \mathcal{F}_{ik}^{\text{model}})$ under the frozen model and its nested evidence snapshots. The expectation in $\mathbb{E}[C_i] = 0$ is over the market’s possible evidence and outcome process, not over decoding randomness. Markets may have different processes; under the null each still has mean-zero C_i , so their topic-weighted population mean is zero. Fresh calls evaluate the same frozen mapping at successive information sets. The bootstrap treats the observed markets as a topic-stratified sample from this target population.

Markets and aggregation. The main analysis contains 73 resolved binary markets across 21 topics. Each model forecasts at 12 common cut-points, using a fresh call and all headlines available by that time. The primary panel contains eight production systems from three providers. The open-weight gpt-oss-120b anchor is reported separately and is

not included in the primary aggregate; two additional efficient systems are held out. Equation (2) weights the eight primary systems equally within market. Averaging within provider before combining the three provider means gives +0.058 ([+0.016, +0.101]). Across the primary systems and anchor, 12 of 657 model–market paths have extraction failures; the estimator adjusts J_i market by market.

Results

A positive model–crowd gap. The mean paired gap is $\Delta C = +0.061$. The topic-clustered interval is [+0.018, +0.105], the market-bootstrap interval is [+0.024, +0.099], and the paired Wilcoxon signed-rank test gives $p = 3.7 \times 10^{-6}$. The primary interval comes from 10,000 two-level bootstrap draws that resample topics, then markets within topics, while retaining every available model path for a sampled market.

Stochastic elicitation stress test. Primary calls use deterministic decoding (temperature 0, top- $p = 1$). To stress-test sampling variability, we repeated the same prompts five times on 15 markets using temperature 0.3 and top- $p = 0.95$. Because these are deliberately more stochastic settings, this is a stress adjustment rather than an estimate of temperature-0 noise. Subtracting the additional stochastic movement reduces the gap by about 15% to +0.052 ([+0.011, +0.094]), which remains positive. A deterministic no-new-evidence placebo contributes movement of only 0.0014 per path. The unadjusted +0.061 remains the primary estimate.

Prospective transfer. As of September 15, 2026, 41 of 52 prospectively frozen markets have resolved, eight remain open, and three became invalid. The resolved-market gap is +0.054 ([+0.012, +0.097], $p = 0.0023$). A later 30-market cohort gives an outcome-free gap of +0.049 ([+0.012, +0.087]). Within the original 52-market cohort, a 28-market subset that remained unresolved throughout elicitation gives +0.058 ([+0.019, +0.098]). These rows provide different checks but are not all statistically independent.

Capability pattern. The gap varies substantially across systems. The eight primary-system marginal means average +0.0635; the pre-specified market-level estimator is +0.061 because missing paths change the available denominator J_i on a small number of markets. The open-weight anchor is excluded from both values and shown separately in Figure 1. A hierarchical analysis gives a slope of -0.024 per pre-assigned capability tier (95% credible interval $[-0.041, -0.008]$), with the same direction within each provider. Tiers use an external capability snapshot frozen on July 15, 2026 [22]. Two held-out efficient systems have gaps of +0.112 and +0.138. We treat this as heterogeneity in a fixed panel, not a universal scaling law.

Robustness and interpretation. The estimate stays positive across observation grids, prompt wording, probability formats, context handling, missing-path treatments, and stricter post-cutoff subsets. Showing the previous crowd probability

reduces the gap to +0.032 ([+0.004, +0.061]), consistent with a stabilizing market signal. Models and traders still have different information sets, so these results identify a robust path difference rather than its complete causal mechanism.

Relation to recent work

The closest martingale studies use different processes and estimands. Falck et al. [9] test exchangeable in-context learning; He et al. [10] regress future belief changes on current beliefs across reasoning trajectories; and Kim et al. [11] analyze repeated-answer drift and stabilization in multiple-choice QA. We instead use the Augenblick–Rabin movement-versus-uncertainty identity on real, dated forecast paths, add a same-market crowd comparator, and test prospective transfer.

Paleka et al. [12] and Andrews and Sarkar [13] test whether sets of probabilistic forecasts admit logical, arbitrage, or Dutch-book violations without requiring outcomes. Those are cross-question coherence tests. Our audit targets within-question temporal coherence: whether movement along one evidence path matches the uncertainty removed from that same path.

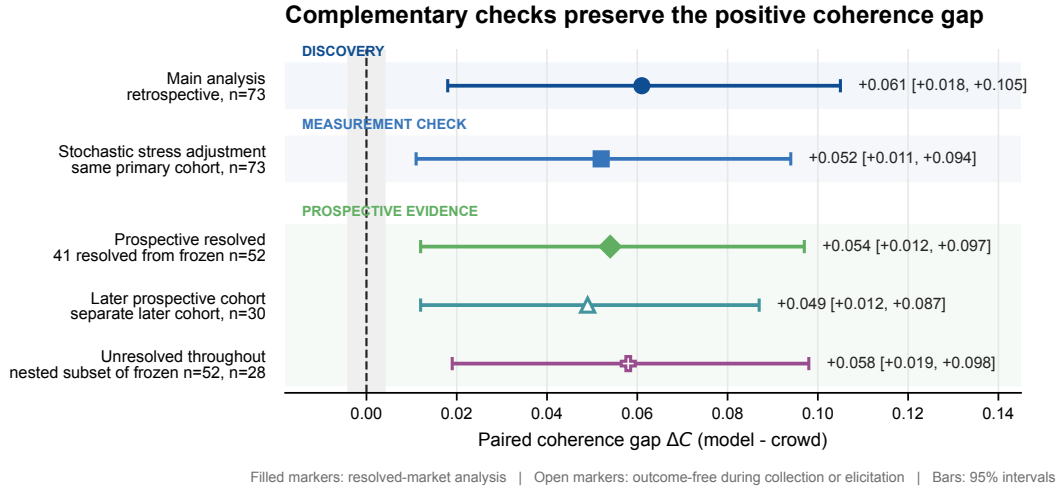
EvolveCast finds that LLM forecasts often under-respond to new information relative to human forecasters [26]. That need not conflict with our larger squared movement relative to a prediction-market crowd. EvolveCast studies update alignment against a human reference; we study a second-moment property accumulated over a path. A forecaster can under-react on typical updates yet accumulate excess squared movement through noisy or occasional large revisions. Comparator and evidence format also differ.

Discussion

What the score adds. Endpoint scores collapse a trajectory to its final probability. The residual instead asks whether total quadratic movement is commensurate with the uncertainty removed between the path’s own endpoints. Two forecasts can therefore finish at the same value and receive the same Brier score while having sharply different C . This distinction matters whenever a probability triggers actions before resolution, such as escalation, hedging, resource allocation, or human review. Large reversals can repeatedly cross a decision threshold even when the final forecast is accurate.

What the result establishes. The strongest evidence is the market-level paired result and its prospective replication. Both use the same estimand and preserve the market as the independent unit. The capability analysis adds interpretation by locating the gap in several efficient systems, but it is not needed for the primary claim. This keeps the conclusion independent of any particular tier boundary.

Limits. The crowd is a useful benchmark, not a Bayesian oracle. Traders may use private signals, price dynamics, and richer news than the model, so ΔC is not a causal estimate under identical information. The model panel is fixed, the markets come from one platform, and topic coverage is uneven. The capability gradient should therefore be tested on more model



The coherence gap is concentrated in efficient systems

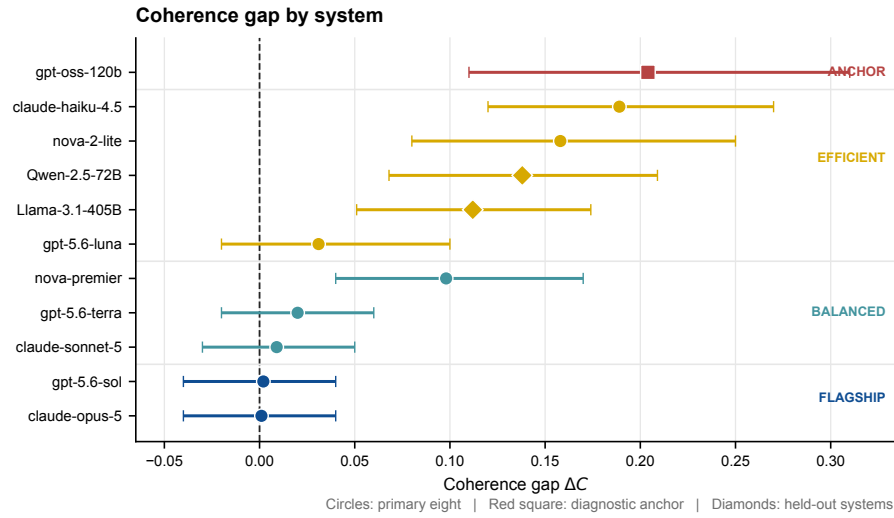


Figure 1: The positive coherence gap replicates and varies by system. Top: Complementary checks distinguish the primary cohort, stochastic stress adjustment, frozen prospective cohort, later cohort, and nested unresolved-at-elicitation subset. Bars are 95% intervals with markets as paired units. **Bottom:** Circles are the eight primary systems, the red square is the separately reported open-weight anchor, and diamonds are two held-out efficient systems. Tiers were assigned before coherence analysis; the primary-panel slope is -0.024 per tier ($[-0.041, -0.008]$).

families and forecasting domains. Finally, the stochastic repetition experiment is intentionally more variable than the primary setting. Its subtraction is informative only as a deliberately harsh stress calculation, not as an identified decomposition of primary-path movement.

Conclusion

Across this fixed panel and market platform, model paths have a positive crowd-relative coherence residual that survives a stochastic elicitation stress test and prospective evaluation. The gap is concentrated in efficient systems, while the strongest systems remain close to the crowd. Broader samples are needed before treating this capability gradient as general. Forecasting systems should be evaluated not only by where they finish, but also by how they change their minds.

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Why the identity holds

Let $(p_k)_{k=0}^K$ be a square-integrable martingale and $\Delta p_k = p_k - p_{k-1}$. Orthogonality of increments gives

$$\mathbb{E}\left[\sum_{k=1}^K (\Delta p_k)^2\right] = \mathbb{E}[p_K^2 - p_0^2].$$

Because $\mathbb{E}[p_K] = \mathbb{E}[p_0]$,

$$\mathbb{E}[p_K^2 - p_0^2] = \mathbb{E}[p_0(1 - p_0) - p_K(1 - p_K)],$$

and therefore $\mathbb{E}[C] = 0$. The equality is in expectation on a fixed grid. Outside the martingale case, refining the grid need not move C in one direction.

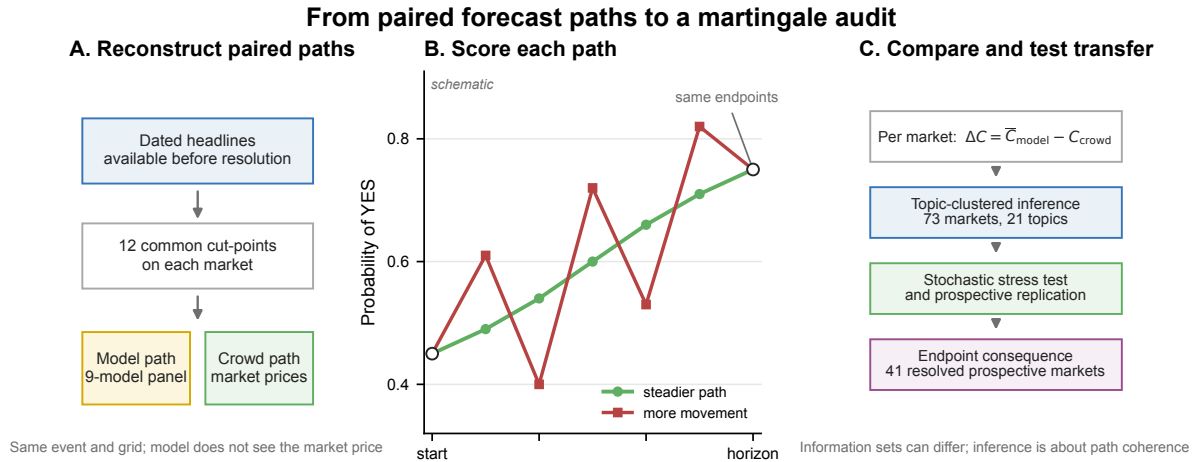
Study details

Setting	Primary forecasts	Repetition sensitivity
Temperature	0	0.3
Top- p	1.0	0.95
Repetitions	1	5
Markets	73	15
Purpose	Main estimate	Stochastic stress test

Table 1: Elicitation settings. Seeds are fixed where the provider supports them; the stress run intentionally introduces more stochasticity and is not treated as an identified correction.

Model	Provider	External score	Tier
gpt-5.6-luna	OpenAI	1198	Efficient
claude-haiku-4.5	Anthropic	1165	Efficient
nova-2-lite	Amazon	1142	Efficient
gpt-5.6-terra	OpenAI	1289	Balanced
claude-sonnet-5	Anthropic	1312	Balanced
nova-premier	Amazon	1301	Balanced
gpt-5.6-sol	OpenAI	1407	Flagship
claude-opus-5	Anthropic	1421	Flagship
gpt-oss-120b	Open weight	N/A	Anchor

Table 2: Tier assignments from the July 15, 2026 external capability snapshot. Boundaries are < 1250 , 1250 – 1350 , and > 1350 ; the open-weight anchor is reported separately and excluded from the primary aggregate.



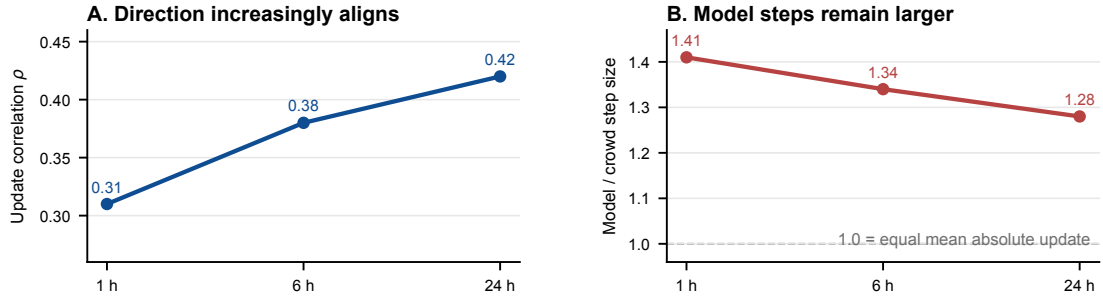
$$M = \sum (\Delta p)^2 \quad R = p_0(1 - p_0) - p_K(1 - p_K) \quad C = M - R$$

Larger movement with the same endpoints gives a larger residual C

Figure 2: Audit workflow and score anatomy. Evidence snapshots are nested, inference is paired by market, and uncertainty is clustered by topic.

Additional diagnostics

Headline-aligned updates: descriptive point estimates (n = 784)



Model and crowd information sets differ; these panels are descriptive.

Figure 3: Headline-aligned descriptive diagnostics. Across 784 headline–market pairs, model and crowd changes increasingly agree in direction over wider windows, while model steps remain larger. These are point estimates, not identical-information causal comparisons.

Analysis	n	ΔC	95% CI
Main topic bootstrap	73	+0.061	[+0.018, +0.105]
Market bootstrap	73	+0.061	[+0.024, +0.099]
Leave-one-topic check	73	+0.061	[+0.021, +0.101]
Stochastic stress adjustment	73	+0.052	[+0.011, +0.094]
Provider-balanced primary	73	+0.058	[+0.016, +0.101]
Similar starting probabilities	51	+0.055	[+0.017, +0.094]
Evidence after model cutoff	58	+0.064	[+0.023, +0.106]
Prospective resolved markets	41	+0.054	[+0.012, +0.097]
Unresolved throughout elicitation	28	+0.058	[+0.019, +0.098]

Table 3: Main estimate and interpretation-relevant checks.

Further checks. The score is stable with 8, 12, or 24 observations per path and changes little across prompt paraphrases, decimal or percentage outputs, logit elicitation, and fresh or accumulated context. Synthetic experiments recover a residual near zero for a Bayesian agent, negative residuals for under-reaction, and positive residuals for over-reaction and noisy updating; interval coverage ranges from 94.6% to 95.2%.